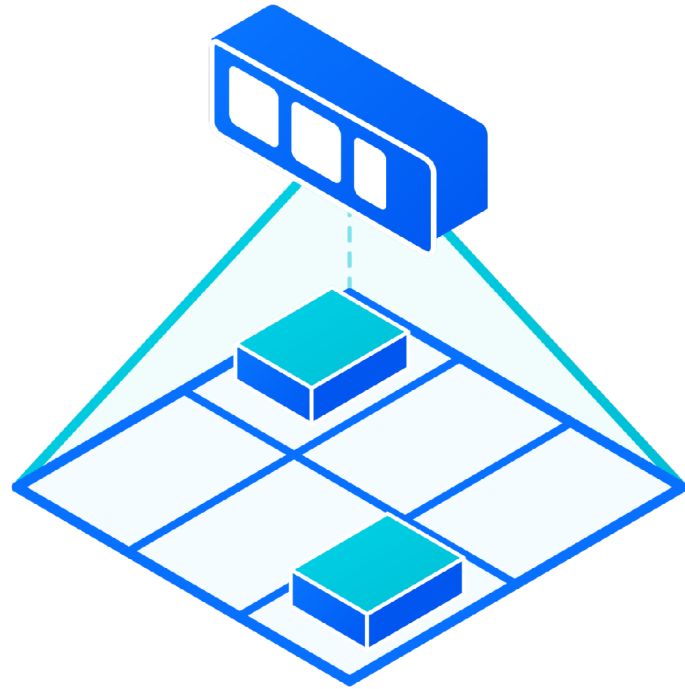




AW3 StockSync

Storage Slot Detection User Manual

Detect slot occupancy, classify cargo and check placement

Contents

1 Product Overview	3
1.1 Install and Connect	3
1.2 Initial Setup	3
Install the Algorithm on the Camera	4
2 Before You Start	5
2.1 Recommended Operating Conditions	6
2.2 Algorithm Limits	6
2.3 Result Definitions and Acceptance	6
3 Run Your First Detection	7
3.1 Main Interface	7
3.2 Coordinates and Extrinsic	8
3.3 Ground Calibration	9
3.4 Set Detection Areas	10
3.5 Set Detection Parameters	12
3.6 Monitor and Verify Results	13
3.7 Daily Operation	15
4 Functions	16
4.1 Volume Measurement and Related Functions	16
4.2 Set XYZ Bounds for Multiple Slots	18
4.3 Preview All	19
4.4 Semantic Recognition	19
5 Communication and Result Data	20
5.1 Configure and Verify Communication	20
5.2 Read Result Messages	20
6 Troubleshooting and Maintenance	22
6.1 System and Streaming Issues	22
6.2 Detection Issues	22
6.3 Export Troubleshooting Data	23
6.4 Maintenance and Rechecking	23
Revision History and Technical Support	24
Technical Support	24

1 Product Overview

StockSync runs in AW3 for slot inspections, inventory checks and checks before picking or placing cargo. It reports empty/occupied status and, when enabled, placement quality, dimensions, coverage and cargo classes.

AW3 manages device connections, camera groups, communication and logs. StockSync handles calibration, slot setup, detection and result display.

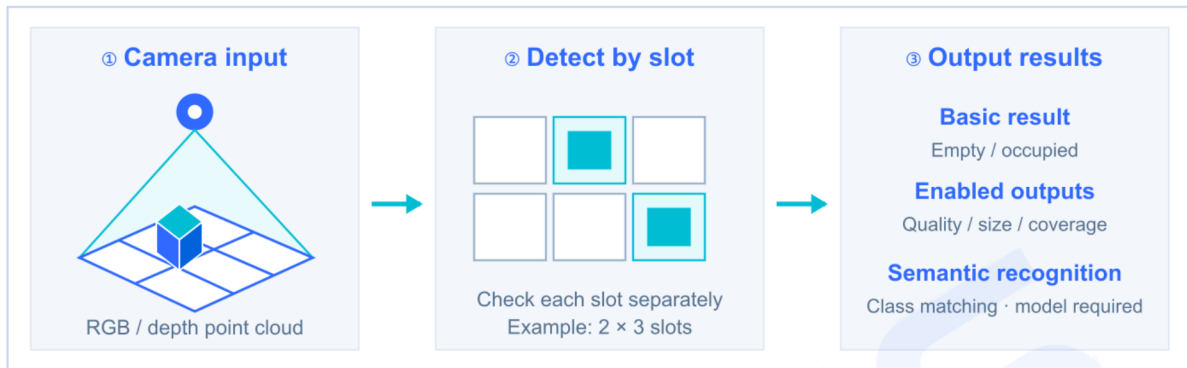


Fig. 1 Storage slot detection workflow

1.1 Install and Connect

We provide two installation packages:

AW3 installer

Install on your computer to connect to and operate the camera

Algorithm package (.tar.gz)

Install on the S10 Pro camera to run StockSync



Network setup Your computer and camera must be on the same subnet, with different IP addresses. Example: computer 192.168.1.10, camera 192.168.1.20, both using subnet mask 255.255.255.0. Use the addresses assigned at your site.

Before you start Deselect the AW3 backend download option when installing AW3. Have the camera IP address, username and password supplied with your system ready. Log in via SSH, then follow page 4 to install the algorithm package.

1.2 Initial Setup



Use [Expert] to configure detection and [Monitor] to view results. Cameras detect independently. Each camera has separate 2D and 3D slot settings.

Install the Algorithm on the Camera

After connecting to the camera, install the supplied algorithm package (.tar.gz) via SSH.

Replace USER and CAMERA_IP with the camera username and IP address. Replace PACKAGE.tar.gz and FOLDER with the actual package filename and extracted folder name.

On your computer: run from the package folder

```
scp "PACKAGE.tar.gz" USER@CAMERA_IP:~/  
ssh USER@CAMERA_IP
```

After logging in via SSH: run on the camera

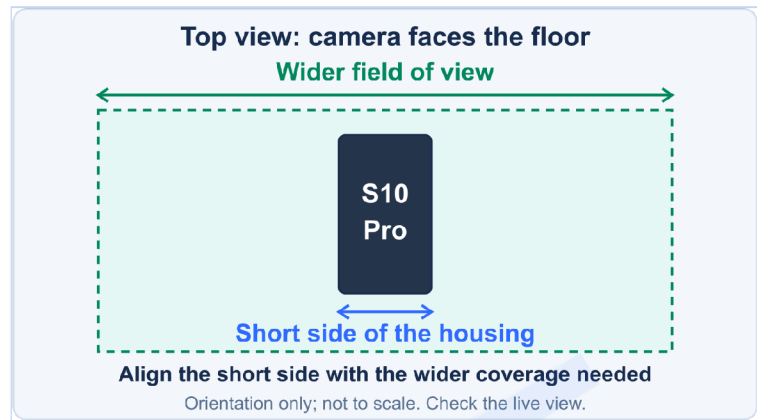
```
cd ~  
tar -xzf "PACKAGE.tar.gz"  
cd "FOLDER"  
sudo ./install.sh  
sudo systemctl status aw3 --no-pager
```

When the status shows active (running), return to AW3 and check that the images and algorithm status are normal.

2 Before You Start

Standard Camera: S10 Pro

The S10 Pro RGB-D camera provides color images and depth data for slot detection.



S10 Pro (left) and coverage orientation (right)

Key specification	S10 Pro
Output	Depth, RGB and IR amplitude images
Depth resolution / rate	240 × 160; maximum and typical rate: 10 fps
RGB resolution / rate	1920 × 1080; maximum and typical rate: 10 fps
Field of view (H × V)	ToF and RGB: 61° × 90° ± 3°
Ranging distance	0.3–17 m (90% reflectivity); 0.3–13 m (10% reflectivity)
Ranging accuracy	≤ 3 cm
Power / average consumption	24 V; ≤ 7 W
Interfaces / synchronization	Ethernet, IO; PTP time synchronization
Dimensions / weight	92 × 47 × 51 mm; 460 g
IP rating / temperature	IP67; −20 °C to 60 °C
Wavelength / eye safety	940 nm; CLASS I

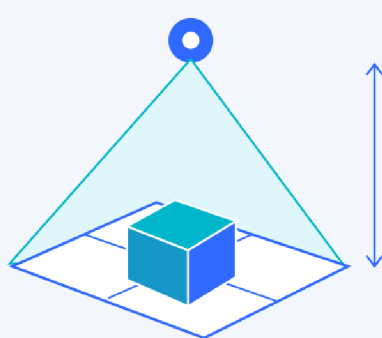
Mounting orientation The S10 Pro has a wider field of view along the short side of its housing. Do not copy the mounting orientation of other MRDVS models. Align this side with the direction requiring wider coverage and confirm it in the AW3 live view. Generic camera symbols do not indicate mounting orientation. Camera specifications are not algorithm limits. See Sections 2.1–2.3 for installation and detection requirements.

Source: MRDVS S10 Series RGBD Camera Product Specification, V2.1. Refer to the 3D model for product dimensions.

2.1 Recommended Operating Conditions

Use the following conditions and verify detection on site.

Fixed camera · full slot coverage



Level surface · calibrate ground first

Installation and environment
Secure S10 Pro; align with the slot layout.
Low dust, no rain; indoor or normal outdoor light.

Installation height
Occupancy detection: up to 8 m.
Volume/quality: target 5 m limit; unverified.

Cargo dimensions
Each dimension ≥ 400 mm; validate at installed height.
40 cm at 5 m: unverified. Test minimum sizes on site.

View and point cloud
Keep slots/cargo fully in view; avoid occlusion/stacking.
Continuous surface points; stable, level support.

2.2 Algorithm Limits

Black objects



Test actual samples

Occluded cargo



Missed detections or incomplete dimensions

Outside the area



Objects may go undetected

Highly reflective objects



Not supported

Fig. 2 Supported materials and detection limits

Materials Test black samples; very dark or low-reflectivity objects may be missed or unmeasurable. Highly reflective objects are unsupported.

Site checks Check that images and point clouds update continuously. Remove obstructions and moving interference; avoid overlapping slots and double counting.

2.3 Result Definitions and Acceptance

 **1 Check slots**

Verify empty and occupied
Match each slot name
to its physical location

 **2 Verify functions**

Edges · angles · multiple objects
Use known-size reference cargo
to check measured dimensions

 **3 Site acceptance**

Use site hardware and mounting
Check cargo and detection cycle
Accept based on on-site results

Angle checks apply to rectangular or palletized cargo with stable, straight edges.

Results "Occupied" means cargo is present, not that placement or class is correct. "—" means invalid or not applicable; it does not mean zero or empty.

Recalibrate and check all slots if the camera position, camera angle or support surface changes.

3 Run Your First Detection

3.1 Main Interface

Select the camera on the left. Check its connection and live views before use.

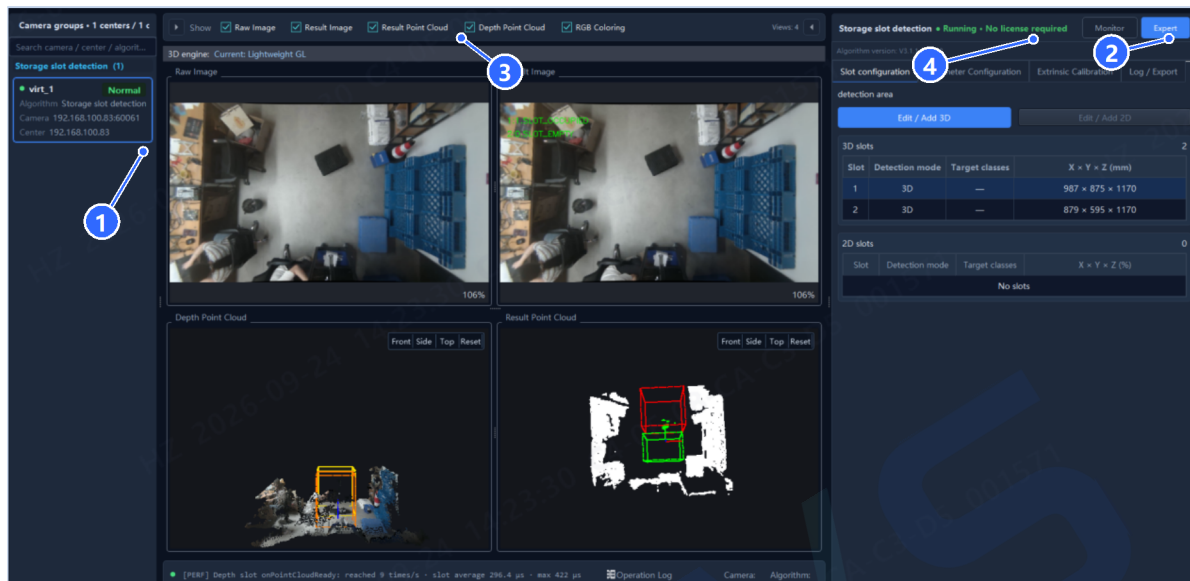


Fig. 3 ①Camera list ②Mode ③View controls ④Algorithm status

Interface area	Function
① Camera list (left)	Select a camera and check its connection. Settings apply to the selected camera.
② Mode (right)	Use [Expert] for calibration, slots and parameters; use [Monitor] for routine result checks.
③ View controls (center)	Select Raw Image, Result Image, Depth Point Cloud or Result Point Cloud. [RGB Coloring] adds scene colors to point clouds without changing slot bounds.
④ Status (top right)	Check algorithm status. For errors, check the connection, live data and whether settings were sent successfully.

3.2 Coordinates and Extrinsics

Open [Expert] → [Extrinsic Calibration]. Use the default ground frame (Z up), then calibrate the ground before defining slots.

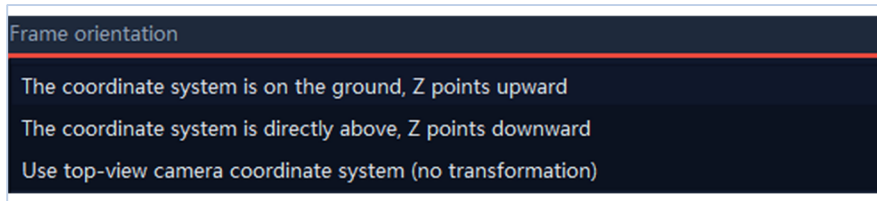


Fig. 4 Coordinate frame options

Option	Use
Ground coordinates, Z upward	Default. Measure height above the calibrated floor or tabletop.
Overhead coordinates, Z downward	Use the axes required by your project. Do not reuse values from a different orientation.
Top-view camera coordinates (no transformation)	Use camera coordinates directly. A person familiar with coordinate transforms should set and verify the axes.

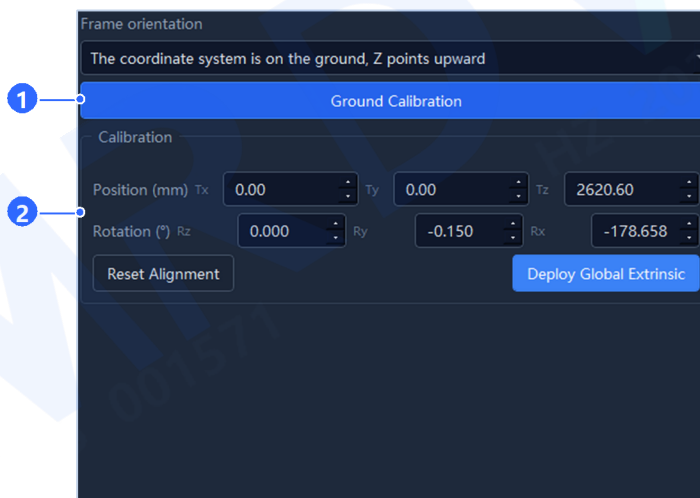


Fig. 5 Extrinsics: ①Ground Calibration ②Extrinsic parameters

Tx, Ty and Tz are in mm; Rx, Ry and Rz are in degrees. After changing coordinates, check the origin, axis directions and slot bounds. [Reset Alignment] does not replace ground calibration.

3.3 Ground Calibration

Calibrate the floor or tabletop supporting the cargo as zero height before creating slots.

Requirements Remove all objects from view to expose a flat, clear floor or tabletop.

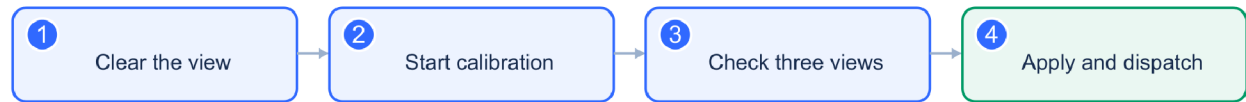


Fig. 6 ①Start calibration ②Three views ③Warnings/results ④Apply and dispatch

Area	Function
① Start calibration	Automatically detect the floor or tabletop.
② Top / front / side	Check for a level plane, with no objects or supports.
③ Warnings/results	Check inlier ratio, tilt and fit. Recalibrate on warnings.
④ Apply and dispatch	Confirm the result, then save and apply the calibration.

Stop and recalibrate For warnings about low inlier ratio, slopes, unevenness or obstructions, clear the area, check the plane and recalibrate. Do not rely on the completion message alone.

Check Confirm in all three views that the support plane is nearly level, the origin lies on the actual surface and calibration was sent successfully.

3.4 Set Detection Areas

Open [Expert] → [Slot configuration] → [Edit / Add 3D] to open the [SlotSpace Calibration and Frame] window.

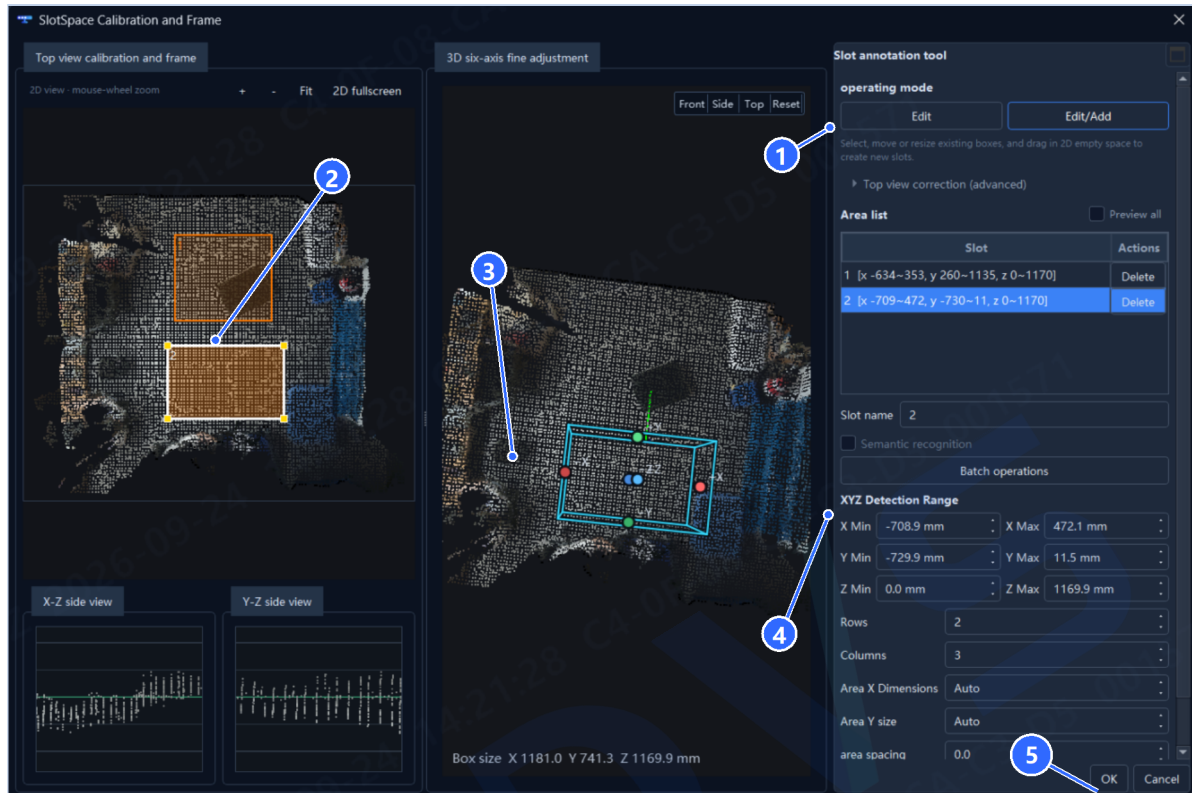


Fig. 7 ①Mode/list ②Top view ③3D check ④XYZ bounds ⑤OK

Step	Action
① Select mode	Use [Edit] to adjust existing slots. Use [Edit/Add] to draw new slots in blank space on the left.
② Adjust XY	Select a slot. Move its box or drag the corners to adjust its X/Y position and bounds.
③ Check 3D	Drag the 3D handles to adjust bounds. Check height and edges in the X-Z and Y-Z views.
④ Name and XYZ	Enter a name. Check X/Y/Z minimum and maximum values to exclude the background and nearby cargo.
⑤ Confirm	Click [OK] to save, apply the settings and return to the main page. [Cancel] discards changes.

Z reference The calibrated surface is $Z = 0$, with Z pointing up. A low Z Min may include the support; a high Z Min may exclude short objects.

Note Use unique slot names/IDs within each camera and mode. Manage 2D and 3D separately. View slots on the main page; edit them in the dialog.

Check Return to the main page and confirm slot count, names and bounds. Test each slot with cargo.

3.4.1 Create Slots in Batches

Create a grid of 3D slots, or add to an existing layout. Check Z bounds first.

1. Scroll down on the right. Set [Rows], [Columns], [Area X Dimensions], [Area Y size] and [area spacing]; click [Enter batch create].
2. Draw, move or resize the grid on the left. Auto calculates slot sizes from the grid bounds.
3. Check the rows, columns, dimensions and spacing. Click [Finish batch create] to confirm or [Cancel batch] to discard the grid.
4. Select [Preview all] to check every 3D box's position, spacing and height. Click [OK] to save and send the settings.

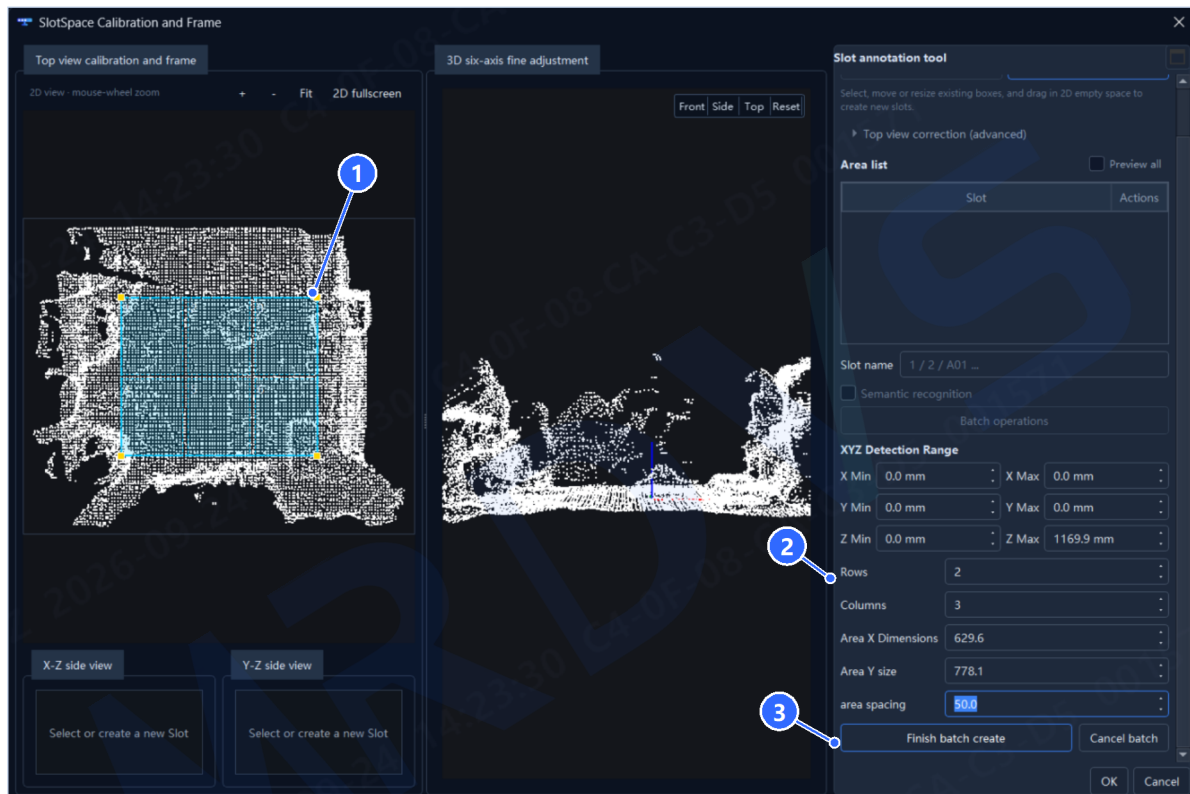


Fig. 8 ①Grid preview ②Rows, columns, size and spacing ③Finish batch create

Example: 2 rows × 3 columns. The cyan grid previews six slots before creation.

Check Confirm slot count, names and positions. Check for unwanted overlap.

3.5 Set Detection Parameters

Open [Expert] → [Parameter Configuration]. Settings apply to all slots on this camera.

1. Adjust one parameter at a time. See Chapter 4 for optional functions.

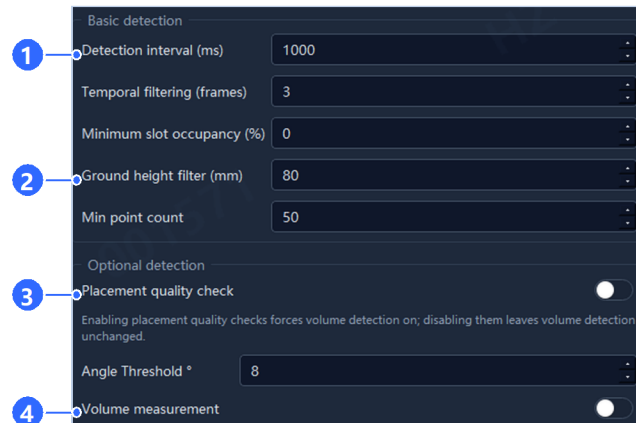


Fig. 9 ①Interval ②Ground filter ③Placement quality ④Volume measurement

Parameter	Adjustment
Detection interval (ms)	Use a shorter interval for faster updates or a longer interval for slower updates.
Temporal filtering (frames)	Increase to stabilize the status. Decrease for a faster response.
Min point count	Increase to filter noise. If small objects are missed, check the point cloud before reducing it.
Ground height filter (mm)	Increase if the ground is detected as cargo. Reduce if low objects are filtered out.
Minimum slot occupancy (%)	Keep the supplied value. Contact your supplier before changing it.

2. Click [Save and send]. Wait for results to stabilize, then check empty and occupied slots.

Before adjusting Check calibration, slot bounds and point clouds. Start with the supplied settings.

3.6 Monitor and Verify Results

In [Monitor], compare the 2D/3D slot tables with the result views and the scene.

Enabled functions The display depends on enabled functions: Volume measurement provides dimensions, Placement quality check provides quality and angle, and Semantic recognition provides classes.

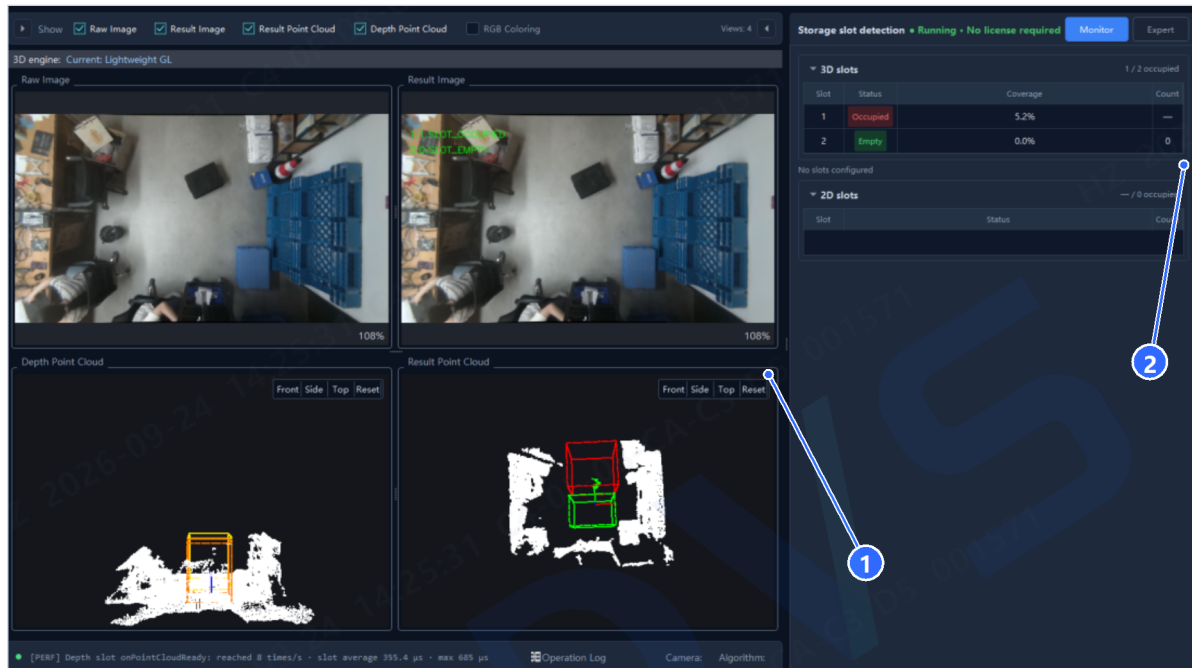


Fig. 10 Monitor: ① Result views ② 3D slot table

Display	Function
Status	Green = empty; red = occupied.
Quality	Shows normal, out of bounds or misaligned when quality checks are enabled.
Coverage	Percentage of slot area covered by cargo.
Count	Number of objects detected in the slot.
Details arrow	Expand or collapse dimensions, classes and angles for enabled functions.

View control	Function
Select views	Select Raw Image, Result Image, Result Point Cloud or Depth Point Cloud.
RGB Coloring	Adds scene colors to point clouds (Figs. 3, 7) without changing slot bounds.
Viewpoint	[Front], [Side], [Top] and [Reset] change only the view.

Check If messages and slot tables disagree, reload the settings. Wait for new results, then check the slot count.

3.6.1 Understand Detection Results

Check slot occupancy first. Then check quality, dimensions and angles for enabled functions.

Result	Meaning	RGB keyword
Empty	No valid cargo.	SLOT_EMPTY
Occupied	Cargo detected.	SLOT_OCCUPIED
Occupied, misaligned	Within bounds, but the angle exceeds the threshold.	NOT_ALIGNED
Occupied, out of bounds	The cargo's bounding rectangle exceeds the allowed boundary.	INTERSECT_WITH_BOARDER

When placement quality checks are enabled, results have three states. "Occupied" alone does not mean cargo is correctly placed.

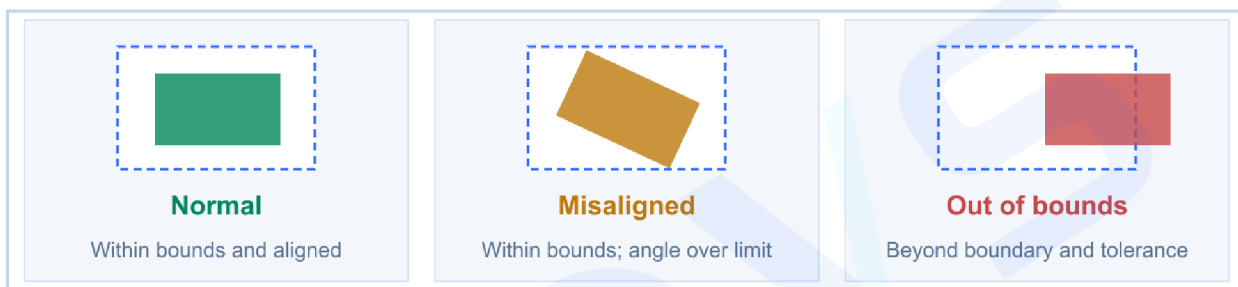


Fig. 11 Placement quality states

Dimensions and Angles

Result	Meaning
Length × width × height	Cargo bounding-box dimensions in mm, including gaps and any part extending beyond the slot.
Angular deviation	Cargo orientation relative to the slot, in degrees. Applies to rectangular or palletized cargo with stable, straight edges. Not checked when cargo is out of bounds.

Measurement limits Missing depth data or excessive ground filtering may cause incomplete measurements.

Invalid results A dash (—) means no valid value, not applicable or function disabled. It does not mean empty. Wait for new results after startup, stream recovery or configuration changes.

3.6.2 Verify Detection

After applying the settings, switch to [Monitor]. Check empty and occupied slots, then test the enabled functions.

Check	Action	Expected result
Empty	Remove cargo and wait for stable status.	The slot shows Empty.
Occupied	Place typical cargo in each slot in turn.	The matching slot shows a stable Occupied status.
Placement quality	Align cargo, rotate past the angle threshold, then extend it beyond the boundary.	Normal, then misaligned, then out of bounds.
Volume measurement	Use cargo of known dimensions. Move part of it outside the slot.	Dimensions meet site requirements and include the part outside. Coverage reflects the area inside.

Test scope Test only enabled functions. Check semantic recognition using the classes supported by the supplied model.

3.7 Daily Operation

1. Start AW3 and select the camera. Check that Raw Image and Depth Point Cloud keep updating.
2. Open [Monitor] and check slot count, positions and occupancy.
3. For out-of-bounds, misaligned or invalid results, check the scene and follow Chapter 6.
4. After changing settings or the installation, repeat the checks in Section 3.6.2.

Check Verify empty and occupied slots and the enabled functions. Keep the settings and measurement records.

4 Functions

4.1 Volume Measurement and Related Functions

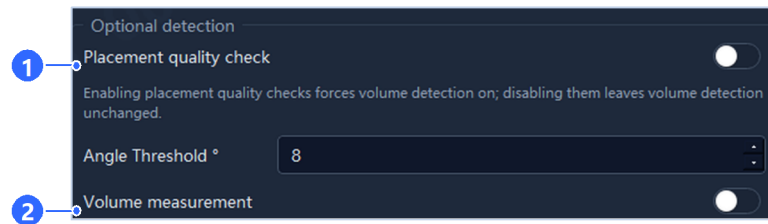


Fig. 12 ①Placement quality ②Volume measurement

Function	Purpose and settings
Volume measurement	Once enabled, expand slot details in [Monitor] to view length, width and height, including parts beyond the slot.
Placement quality check	Classifies placement as normal, misaligned or out of bounds. A lower angle threshold requires closer alignment.

Linked settings Placement quality check also turns on Volume measurement. Turning quality checks off does not turn volume measurement off.

View Measurement Results

Expand slot details in [Monitor] to view each object's dimensions and angle.

Status	Quality	Coverage	Count	Details
Empty	—	0.0%	0	▼
1 Occupied	Out of bounds	36.6%	1	▼
2 L × W × H (mm)		Deviation (°)		
512.1 × 427.9 × 208.6		17.3		

Fig. 13 ①Slot status and quality ②Object dimensions and angles

Out-of-bounds cargo Volume measurement still works when cargo extends beyond the slot. Dimensions include the overhanging portion; coverage counts only the area inside the slot.

4.2 Set XYZ Bounds for Multiple Slots

Use [Batch operations] to apply the same X, Y or Z limits to several slots.



Fig. 14 ①Select slots ②Select coordinates and enter values ③Apply

1. Select the slots to update.
2. Select the minimum or maximum coordinates to change and enter the values to apply.
3. Click [Apply to N slots] and check the bounds. Click [OK] to save and apply the settings.

Height only Select Z Min or Z Max to change height limits. Other coordinates stay unchanged.

4.3 Preview All

Select [Preview all] in [SlotSpace Calibration and Frame] to display every slot in 3D.

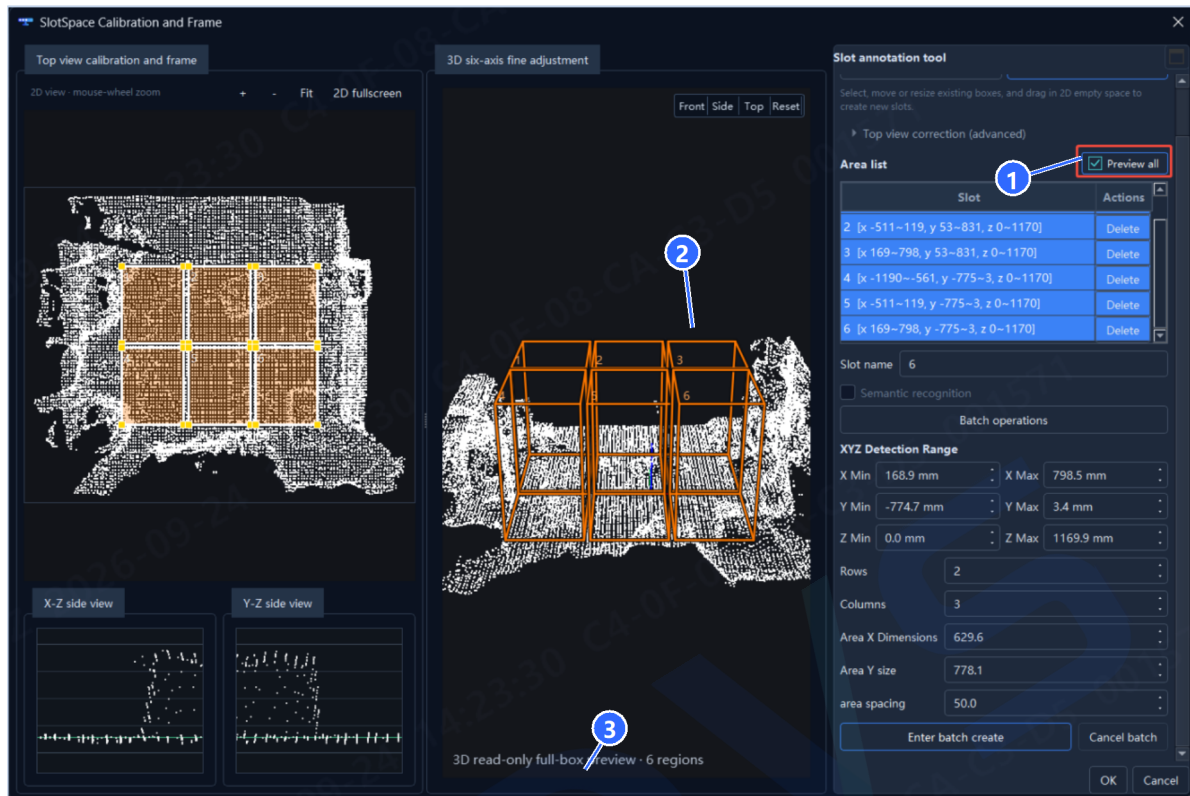


Fig. 15 ①Preview all ②All 3D boxes ③Read-only notice and region count

1. Select [Preview all] to check the positions, gaps and heights of all slots in the central 3D view.
2. Use [Front], [Side] or [Top] to check the layout. The number of regions appears at the bottom.
3. To edit a slot, clear [Preview all], select the slot and adjust its bounds.

Read-only preview Preview does not change slot bounds. After editing, click [OK] to save and apply the settings.

4.4 Semantic Recognition

Semantic recognition identifies cargo classes supported by the model and checks them against each slot's target class.

Availability Semantic recognition is not included in the standard release. Contact your supplier for a model and supported classes.

5 Communication and Result Data

5.1 Configure and Verify Communication

AW3 centrally manages communication. Follow the AW3 Algorithm Platform User Manual to configure incoming triggers and result forwarding.

Test Send a trigger from the receiving system. Check the camera, slot and result.

5.2 Read Result Messages

① Result object

Top-level keys: slot IDs, e.g. A01

② Slot result

type · status · quality · count · coverage_percent

③ Object details and feature flags

semantic · detail__info · is_target · feature flags

Result structure: top-level slot ID → slot result (no data wrapper)

Field	Meaning
type	Matches the configured rule: 0 = 2D semantic; 1 = 3D; 2 = semantic + 3D. Retained for empty slots. Legacy conversion returns null when the type is missing; it is not inferred.
status	0 = empty; 1 = occupied; negative = invalid/unknown. Class mismatch, misalignment and out-of-bounds cargo still mean occupied.
quality	0 = unchecked; 1 = normal; 2 = misaligned; 3 = out of bounds; 4 = class mismatch; 5 = occupancy detected, but no valid measurement details. Quality is separate from occupancy; is_target reports class matching.
count	Object count from volume measurement, including separate objects of the same class. Without volume measurement, occupied slots return null (unknown count); empty slots return 0.
coverage_percent	Range: 0–100. 3D: proportion of occupied XY grid cells. 2D: proportion of recognized pixels within the RGB slot region. The two measures are not directly comparable.

Reading results Read status first, then interpret details using quality and the feature flags. null is not 0; class indices are not object counts.

5.2.1 JSON Result Example

Slot A01 uses semantic + 3D detection. It is occupied with normal quality. Both objects match class 1; volume measurement and alignment checks are enabled.

```
{
  "A01": {
    "type": 2, "status": 1, "quality": 1,
    "count": 2, "coverage_percent": 65.2,
    "is_target": 1, "semantic": [[1], [1]],
    "detail_info": [
      [296, 258, 261, 1100, 600, 204, 0],
      [1600, 258, 261, 400, 300, 204, 0]
    ],
    "volume_detection": true,
    "alignment_checked": true
  }
}
```

Example notes [[1], [1]] means two objects of the same class, so count is 2. The fields below supplement the preceding page.

Field	Meaning
is_target	-1 = not identified/not configured; 0 = no target-class match; 1 = all objects match; 2 = some match, including cases where other objects have unknown classes.
semantic[i]	Class-index array for object i; it may be empty or contain multiple indices. Indices are not object counts. With volume measurement off, only slot-level classes are summarized; occupied slots still return count: null.
detail_info[i]	[center X, Y, Z, length, width, height, angle]. Coordinates and dimensions: mm; angle: degrees, or null when alignment checks are off. With volume measurement on, entries correspond one-to-one with semantic.
volume_detection / alignment_checked	Whether volume measurement/alignment checks are enabled for this slot. Neither runs for 2D slots.
ret_code / error_str	Failures still include the return code and error message. Successful results exclude legacy detail keys formed by appending suffixes to the slot ID.

6 Troubleshooting and Maintenance

Check that Raw Image and Depth Point Cloud keep updating, then troubleshoot below.

Condition	Go to
No image, frozen image or camera offline	6.1 System and Streaming Issues
Live data, but incorrect status, dimensions or count	6.2 Detection Issues

Stop using results During a fault, do not rely on frozen images, old values or unverified results from external systems.

6.1 System and Streaming Issues

Symptom	Action
Camera offline / no image	Check power, the network cable, camera groups and the AW3 connection.
Stream interrupted / frozen	Check camera and central service connections. After recovery, check that Raw Image and Depth Point Cloud update continuously.
Stream fails after camera utility use	If streaming fails after using LxCameraViewer, select [Device Connection / Base Camera] → [Central Service] → [Restart current center].

6.2 Detection Issues

Symptom	Action
Algorithm error	Check that area and parameter settings were applied. If the error persists, check the logs.
Incorrect occupancy	Check point clouds, slot bounds, ground filtering and Min point count.
Unstable status	Clear obstructions and moving objects. Check slot edges, then adjust Temporal filtering.
Incorrect placement quality	Check coordinate orientation, the angle threshold, cargo bounds and Placement quality check.
Dimensions too small / incorrect height	Check point clouds, ground filtering and the zero level.
Incorrect count	Check object spacing and overlapping slots.
Ground calibration warning	Clear the view and expose a flat support surface. Eliminate slopes and obstructions, then recalibrate.

Communication If AW3 shows results but the receiving system gets none, check forwarding, address, port and the receiving service. See Chapter 5.

6.3 Export Troubleshooting Data

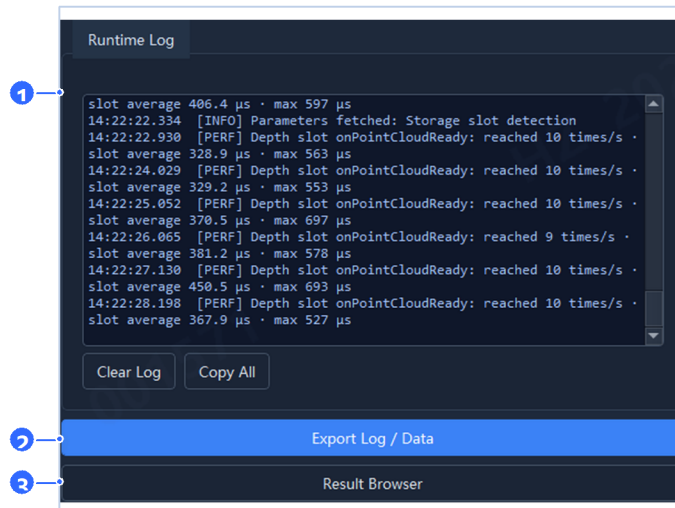


Fig. 16 ①Runtime Log ②Export Log / Data ③Result Browser

1. Open [Expert] → [Log / Export]. Record the time and error message. Use [Copy All] if you need to copy the log.
2. Click [Export Log / Data], select the algorithm, category and time range, and export.
3. Use [Result Browser] to find the matching result images and JSON. Export logs before clearing them.

For support Include versions, camera and slot details, settings, site photos, incorrect results and steps to reproduce the issue.

6.4 Maintenance and Rechecking

Change	Required action
Camera position or angle	Recalibrate the ground and check all slot bounds.
Floor or tabletop height	Recalibrate the ground and check slot Z bounds.
Software, algorithm or parameters	Back up settings. Follow the technical support procedure when redeploying or reinstalling an existing system. Then recheck results and cycle time with representative cargo.
Routine maintenance	Check the lens, mounts and cables. Follow the hardware cleaning instructions. Keep the calibrated camera in place.

Before resuming Back up settings and calibration before making changes. After maintenance, recheck occupancy, placement quality and enabled functions. See Section 3.6.

Revision History and Technical Support

Version	Date	Revision
V0.1	2026-09-24	Added setup, calibration, slot configuration, parameters, monitoring, communication and troubleshooting.

Technical Support

Zhejiang MRDVS Technology Co., Ltd.

Hotline: +86 400-025-6680

Website: www.mrdvs.com

GitHub: <https://github.com/Lanxin-MRDVS>

Knowledge Community: <https://docs.mrdvs.com/en>

LinkedIn: <https://www.linkedin.com/company/mrdvs>



Scan to open MRDVS on GitHub